

Vector Differentiation: Orthogonal Surfaces

1. Orthogonality of Surfaces

Name of Concept

Orthogonal Surfaces (Perpendicular Surfaces).

Core Mathematical Formula

Two surfaces are orthogonal (perpendicular to each other) at a given point of intersection if their respective normal vector gradients (∇u and ∇v) are perpendicular. This condition is satisfied when their vector dot product equals zero:

$$\nabla u \cdot \nabla v = 0$$

Beginner-Friendly Explanation of Operations

- Orthogonal Means 90° :** In vector calculus, the word "orthogonal" simply means perpendicular or intersecting at a right angle.
 - Gradients as Normals:** For any surface equation written in the form $u(x, y, z) = 0$, the gradient ∇u gives a vector that points straight out of the surface. This is its normal vector.
 - The Dot Product Condition:** If two surfaces meet at a 90° angle, their normal vectors must also meet at a 90° angle. The dot product of any two perpendicular vectors is always zero because $\cos(90^\circ) = 0$. We use this algebraic property to set up an equation and solve for unknown constants.
 - Point of Intersection Constraint:** Since the given point lies on both surfaces, the coordinates of the point must satisfy the equations of both surfaces. This provides an additional algebraic equation to find our unknowns.
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2. Step-by-Step Problem Solution

Question

Find the constants a and b so that the surface $ax^2 - byz = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$.

Step 1: Define the functions by moving all terms to one side

Let the first surface be u and the second surface be v :

$$u = ax^2 - byz - (a + 2)x$$

$$v = 4x^2y + z^3 - 4$$

Step 2: Utilize the point constraint on the first surface

Because the point $(1, -1, 2)$ lies directly on the surface u , substituting $x = 1$, $y = -1$, and $z = 2$ into the equation of u must equal 0:

$$a(1)^2 - b(-1)(2) - (a + 2)(1) = 0$$

$$a + 2b - a - 2 = 0$$

$$2b - 2 = 0$$

$$2b = 2$$

$$b = 1$$

Step 3: Set up and calculate the gradient for the first surface u

The gradient vector field formula is:

$$\nabla u = \bar{i} \frac{\partial u}{\partial x} + \bar{j} \frac{\partial u}{\partial y} + \bar{k} \frac{\partial u}{\partial z}$$

Compute the partial derivatives of u :

$$\frac{\partial u}{\partial x} = 2ax - (a + 2) = 2ax - a - 2$$

$$\frac{\partial u}{\partial y} = -bz$$

$$\frac{\partial u}{\partial z} = -by$$

Combine them into the gradient vector:

$$\nabla u = \bar{i}(2ax - a - 2) + \bar{j}(-bz) + \bar{k}(-by)$$

Step 4: Evaluate ∇u at the point $(1, -1, 2)$

Substitute $x = 1$, $y = -1$, and $z = 2$ into our gradient expression to get the normal vector $\bar{a}$:

$$\bar{a} = \nabla u_{(1,-1,2)} = \bar{i}(2a(1) - a - 2) + \bar{j}(-b(2)) + \bar{k}(-b(-1))$$

$$\bar{a} = \bar{i}(a - 2) + \bar{j}(-2b) + \bar{k}(b)$$

Step 5: Set up and calculate the gradient for the second surface v

The gradient vector field formula is:

$$\nabla v = \bar{i} \frac{\partial v}{\partial x} + \bar{j} \frac{\partial v}{\partial y} + \bar{k} \frac{\partial v}{\partial z}$$

Compute the partial derivatives of v :

$$\frac{\partial v}{\partial x} = 8xy$$

$$\frac{\partial v}{\partial y} = 4x^2$$

$$\frac{\partial v}{\partial z} = 3z^2$$

Combine them into the gradient vector:

$$\nabla v = \bar{i}(8xy) + \bar{j}(4x^2) + \bar{k}(3z^2)$$

Step 6: Evaluate ∇v at the point $(1, -1, 2)$

Substitute $x = 1$, $y = -1$, and $z = 2$ into our gradient expression to get the normal vector $\bar{b}$:

$$\bar{b} = \nabla v_{(1,-1,2)} = \bar{i}(8(1)(-1)) + \bar{j}(4(1)^2) + \bar{k}(3(2)^2)$$

$$\bar{b} = -8\bar{i} + 4\bar{j} + 12\bar{k}$$

Step 7: Apply the orthogonality condition ($\bar{a} \cdot \bar{b} = 0$)

Multiply corresponding components ($\bar{i}$ with $\bar{i}$, $\bar{j}$ with $\bar{j}$, $\bar{k}$ with $\bar{k}$) and sum them up:

$$(a - 2)(-8) + (-2b)(4) + (b)(12) = 0$$

$$-8a + 16 - 8b + 12b = 0$$

$$-8a + 16 + 4b = 0$$

Step 8: Substitute the value of b to solve for a

From Step 2, we already know that $b = 1$. Substitute this value into our orthogonality equation:

$$-8a + 16 + 4(1) = 0$$

$$-8a + 20 = 0$$

$$8a = 20$$

$$a = \frac{20}{8}$$

$$a = \frac{5}{2}$$

Final Answer

The required constant values are:

$$a = \frac{5}{2}$$

$$b = 1$$